Homework #1 Solutions

① Construct a formal proof of the following claim.

Claim: The set $P = \{x \in \mathbb{R}^n \mid Ax = b,\ Cx \leq d\}$ is a convex set in $\mathbb{R}^n$, where A, b, C, d are appropriately-sized real matrices/vectors.

Proof: let $P = \{x \in \mathbb{R}^n \mid Ax = b,\ Cx \leq d\}$ for given matrices/vectors A, b, C, d. let $x, y \in P$ and $z = \lambda x + (1-\lambda) y$ for $0 \leq \lambda \leq 1$. We want to show $Az = b$ and $Cz \leq d$ for then $z \in P$ and thus P is convex. We will use the fact that $Ax = b$, $Ay = b$, $Cx \leq d$ and $Cy \leq d$.

$$Az = A(\lambda x + (1-\lambda) y) = \lambda Ax + (1-\lambda) Ay = \lambda b + (1-\lambda) b = b.$$

$$Cz = C(\lambda x + (1-\lambda) y) = \lambda Cx + (1-\lambda) Cy \leq \lambda d + (1-\lambda) d = d.$$

$\blacksquare$

Note: We just proved that the feasible region of a linear program is a convex set.

② Suppose that a function $f: \mathbb{R}^n \to \mathbb{R}$ is both concave and convex. Prove that f is an affine function.

Proof: Suppose $f: \mathbb{R}^n \to \mathbb{R}$ is both concave and convex. (consider f smooth). Since f is convex, $f(y) \geq f(x) + (y-x)^T \nabla f(x)$ and since f is concave, $f(y) \leq f(x) + (y-x)^T \nabla f(x)$.

Thus, $f(y) = f(x) + (y-x)^T \nabla f(x) \quad \forall x, y \in \mathbb{R}^n$.

In particular, if we swap x and y we find that $\nabla f(x) = \nabla f(y)$. That is, $\nabla f(x)$ is a constant vector, say $c = \nabla f(x)$.

In other words f is affine with constant gradient. $\blacksquare$

If ∇f does not exist for all $x \in \mathbb{R}^n$, then a different approach is necessary.

③ Suppose that $f_1, \dots, f_m$ are convex functions from $\mathbb{R}^n$ into $\mathbb{R}$ and let $f(x) = f_1(x) + f_2(x) + \cdots + f_m(x)$.

ⓐ Show that if each f_k is convex then f is convex.

Proof: Suppose $f_1, \dots, f_m$ convex functions from $\mathbb{R}^n$ into $\mathbb{R}$. Define $f: \mathbb{R}^n \to \mathbb{R}$ as $f(x) = f_1(x) + \cdots + f_m(x)$. We wish to show that $f(\lambda x + (1-\lambda) y) \leq \lambda f(x) + (1-\lambda) f(y)$ for arbitrary $x, y \in \mathbb{R}^n$ and $0 \leq \lambda \leq 1$.

$$
\begin{aligned}
f(\lambda x + (1-\lambda) y) &= \sum_{k=1}^{m} f_k(\lambda x + (1-\lambda) y) \\
&\leq \sum_{k=1}^{m} \lambda f_k(x) + (1-\lambda) f_k(y) \\
&= \lambda \sum_{k=1}^{m} f_k(x) + (1-\lambda) \sum_{k=1}^{m} f_k(y) \\
&= \lambda f(x) + (1-\lambda) f(y) \qquad \blacksquare
\end{aligned}
$$

ⓑ Show that if each f_k is piecewise linear and convex, then so is f.

Proof: This takes some notational care and space so I will not compose a complete proof here. One idea is to show

(a) the sum of affine functions is affine

(b) f can be divided into a finite number of regions which are affine.

(c) f can be expressed as the maximum over the affine functions of (b).

(d) We showed in class that such an f is convex.